# Text mining by using Python: Application to patent documents

(2025 Project description)

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1. Introduction

This project is for understanding practices of applying NLP techniques for technology management analysis using patent text information. The project starts with a course work for one week with some preparatory activities, followed by individual project works (mainly topic modeling of some particular type of technology) for some weeks.

2. Setting

- PC with Python: You need to install Python 3.X "anaconda" environment. We use "Jupyter notebook" inside "anaconda" for this project. It can be downloaded freely.
- I will let you access to the shared drive of Google Suite of my laboratory (motolabo.net). Python_Text_Mining holder contains
  - Some files used in this project (including this file)
  - Coursera_video directory: Michigan U, "Applied Text Mining by Python", referred as "VIDEO" here after
  - Data for topic modeling project
- Please check whether you could access to the following holder. If it is not the case, report to prof. Motohashi with your name and e-mail address. https://drive.google.com/drive/folders/1eIbsqmaCnY8tjryYRnriZ2Wu_stQl7g9?usp=sharing

  The folder of "Internahip_files" is only for those who did the phase 2 of this project, " innovation discovery from patent and web contents information"

3. Preparatory activities (Will be introduced at the orientation day)

- Preparatory activities to start course work is provided in the appendix 1.

4. Whole structure

- Part 1 : June 9- June 13 : course works with practical exercise

- Part 2： June 16 – August 1 ：internship project with real data

**[Phase 1]: Learning text mining technique by small sample data： June 9 – June 13 (Orientation will take place on June 2, as follows)**

Venue：Zoom Conference Room：https://u-tokyo-ac-jp.zoom.us/j/99083411670?pwd=U0dicHlONXNiUjBmUFU3Y1h6NWFZdz09

**Orientation Day: June 2, 10:00-12:00 Thai& Vietnam Time (12:00-14:00, JST)**

- Explanation of home works etc (Appendix 1)
- Web crawling introduction by Suchit

**Day1: June 9, 10:00-12:00 Thai& Vietnam Time (12:00-14:00, JST)**

- Review of **preparatory activities**
- How to convert technology contents info (in patent document) to numeric (vector) information?： Keyword extraction, TF-IDF, Word/Document embedding (such as Word2Vec)
- **HOMEWORK1**: Extract three **keywords by using TF-IDF score from 25 AI patents abstracts**, supplied as a home work file. TF-IDF score can be obtained easily by Python module like "scikit-learn" and "genism". But I would advice you to do it without such modules for your understanding of the concept. There are many hints over internet such as https://towardsdatascience.com/text-summarization-using-tf-idf-e64a0644ace3

**Day 2: June 11, 10:00-12:00 Thai& Vietnam Time (12:00-14:00, JST)**

- Review of homework of Day 1
- What are ML applications of patent text information?
- **Preprocessing of text**: tokenization regularization (lowering character), stemming/lemmatization and stop words exclusion
- **Text processing of tf-idf vectors:**

① Create dictionary: mapping every word to a number

② Corpus (list of bags of words)：a list of number of words occurring in each documents

- **HOMEWORK2**: Calculation of **similarity measures** across each documents, and find out the closed documents of each of 25 AI patents ：Now please use genism module, again there are many hints over internet such as "How do I

compare document similarity using Python?" by Jonathan Mugan
https://www.oreilly.com/content/how-do-i-compare-document-similarity-using-python/

**Day 3: June 13 10:00-12:00 Thai& Vietnam Time (12:00-14:00, JST)**

- Review of homework of Day2
- Understanding **technology trend by patent information (clustering vs topic modeling)**
- Watch VIDEO 16&17
① Understanding the concept of topic modeling
② LDA model by using genism module
- ~~Introduction for your **project work** : Pick up one of subject areas shown in the appendix 2 to construct LDA topic model, then understand the contents of technological development of that field. More instructions will be provided later with individual consultation of works to be done.~~
- Some references
① There is a very easy to read document of LDA model available in the shared drive. (Beginners_Guide_Topic_Modeling.doc)
② Good reference about topic modeling by genism
https://www.machinelearningplus.com/nlp/topic-modeling-gensim-python/

**[Phase 2]: Project work with final presentation : June 16 – August 1**

Orientation Day: June 16, 10:00-12:00 Thai& Vietnam Time (12:00-14:00, JST)

- Orientation of phase 2
- Dual Attention Model

**(Outline of works, details to be announced)**

- This process is using NLP techniques above to real world exercises, such as technology opportunity discovery tasks. Please refer to the following Youtube video for this research project.
https://www.youtube.com/watch?v=Y6TIHxfKsmM&t=581s
- Major tasks are scraping company's web pages and using such text information together with patent information to predict market opportunities based on particular type of technology.
- Web scraping task will start after June 2 orientation (with a short course of web scraping)
- Some research assistant honorarium will be offered to this works.

# Appendix1 Preparatory Homework

1. Python environment and understanding on basic command

- You need to bring your own PC with Python 3.X anaconda environment, particularly Jupyter notebook (download for free)
- Familiarize yourself with basic Python function dealing with text, such as making word list from sentence, word count and distribution
- Playing with NLTK library of Python

2. Tasks

- Watch **Video 01,05 and 06** in "Coursera_Video" directory) for the following contents
- What is National Language Processing?
- Playing with NLTK (Natural Language Processing Tool Kit)
- Use the patent abstract data (25 AI related patent abstracts, provided as "homework.csv"), to calculate
- How many sentences, words, unique words
- List of words occurring no less than 10 times **and** no less than 5 character lengths
- After regularization (lowering characters), then comparing the case of Porter Stemmer and Lenmatizer. Compare the results and which one would you think it is better and why?
- Any other observations?

3. Notes

- You may go ahead to watching other videos, but I will give you some general directions.
- **VIDEO 02 and 03** are for regular expression, working with Pandas module of python (concept of "data framework")：No need to worry about them for now, but it will be useful for **rule based text manipulations**.
- **VIDEO 04** is for character code information for various languages (Python's default character set is UFT-8, no need to worry about this for now either.
- Watching VIDEO 07 is optional (could be helpful for your subsequent tasks)